

**PALETTE:
AN ENVIRONMENT FOR DEVELOPING AN INDIVIDUALIZED SET
OF PSYCHOPHYSICALLY SCALED TIMBRES**

William L. Martens
Northwestern Computer Music
School of Music
Northwestern University
Evanston, IL 60201

ABSTRACT

Psychophysical methodology is very well suited to the task of developing a palette of timbres for use in computer music composition and production. By asking for direct judgments about subjective relationships within a set of timbres, perceptual values can be mapped onto the synthesis parameter values associated with those timbres. Such a mapping permits the control of synthesis through the use of perceptual terminology rather than through the specification of cumbersome and non-intuitive synthesis parameters. *Palette* is a computer music environment that supports the automatic psychophysical scaling of synthetic timbres. It is designed to allow users to develop a set of individualized timbral descriptors for the desired perceptual dimensions within their timbral palette. The subjective scales for these dimensions are mapped directly into values of synthesis parameters. The *Palette* environment is composed of several interdependent psychophysical tools that conduct short timbral experiments - the results of which are pooled in a database used for the perceptual control of synthesis. The four primary tools are as follows:

- 1) A stimulus selection tool, which allows real-time timbral adjustment through interaction with a graphic representation of synthesis parameters;
- 2) A unidimensional scaling tool, which collects subjective judgments on a single timbral dimension for a set of multidimensional sound stimuli;
- 3) A timbre matching tool, which establishes the synthesis parameter values that produce timbres matching a standard stimulus on a particular timbral dimension;
- 4) A multidimensional scaling tool, which collects perceptual similarity judgments for pairs of stimuli taken from the timbral palette under development.

In an interactive session with these tools, the amount of timbral data grows until synthesis control has all of the subtlety and complexity conceptualized by the user. Furthermore, the results obtained for a given timbral palette will often provide other individuals with a good starting point for new directions of timbral exploration and psychophysical mapping.

1. INTRODUCTION

During the early development of computer music, research on synthesis algorithms focused on different issues than today. In the infancy of computer technology, the primary concern was the computational efficiency of the algorithm. But there was also a concern with which synthesis strategies would produce musical results. For example, a general strategy was to try to match the waveforms of instruments, as published in textbooks on musical acoustics, but these descriptions proved insufficient. It wasn't until Risset and Mathews (1969) demonstrated the importance of time-varying harmonic amplitudes through analysis and resynthesis that computer music turned its attention to the production of realistic timbres. With today's technology, computational efficiency is not nearly as important as it once was. In a real-time synthesis environment, controls that are psychologically relevant are much more important.

This paper describes a software package that provides composers with perceptual control over timbral synthesis. What this means is that the user need not specify timbres in terms of synthesis parameters; rather, timbres need only be specified in terms of their perceptual qualities. The software package has been named *Palette*, since its purpose is to allow for the development of an individualized palette of timbres for use in music production. The *Palette* project is an example of applied research in computer music that makes use of methodology from perceptual psychology and psychophysics. The package is intended to be used expressly toward creative ends in music composition and production. The multifaceted nature of computer music requires that perceptual researchers, audio engineers, and musicians all work together and influence one another; and our work at *Northwestern Computer Music (NCM)* on *Palette* is an attempt to integrate all their contributions into the development of useful musical tools. All of this work is being conducted on a number of high-resolution graphic workstations (from SUN microsystems), which are connected (through a 10-megabit ethernet) to a file-server that controls both soundfile playback (as described in Roth, Decker, and Kendall, 1985) and real-time synthesis via the Musical Instrument Digital Interface (MIDI). We currently support two *Yamaha DX7s* for input and a *TX816* (eight 16-voice *TFIs*), but any MIDI-controlled synthesizer or sound processor may be incorporated into our system.

The convenience of real-time sound synthesis for computer music has always been desired, but real-time control of synthesis is more than just convenient – it opens up new possibilities for the exploration and evaluation of musical material. Beyond the invaluable experimentation made possible through the tweaking of knobs on analog synthesizers that has been largely absent from computer music, the power of the general purpose computer allows for the rapid development of perceptual intelligence within the workstation software itself. The specific type of perceptual intelligence to be incorporated in the musician's working environment is that which encodes information about how a given sound is perceived when placed into a given musical context. Real-time interaction with timbral synthesis facilitates the acquisition of pertinent perceptual data – the psychophysical relations between synthesis parameters and timbre. These psychophysical relations may be determined through the use of a family of experimental procedures called "psychophysical scaling."

2. PSYCHOPHYSICAL SCALING OF SYNTHETIC MUSICAL TIMBRES

The primary goal of psychophysical scaling is the development of explicit quantitative relations between physical measures of a stimulus attribute and subjective measures of the associated perceptual responses for a given set of stimuli. In the development of perceptual controls for synthesizers, the physical stimulus is specified by a number of synthesis parameters, and the perceptual response is, generally speaking, some judgment about the musical timbre that results. For example, a given synthesis algorithm might produce a range of timbres varying along a single perceptual dimension such as brightness. But the synthesis parameter that controls the brightness of that particular voice on the synthesizer does not produce perceptually equal changes in brightness for equal-sized changes in the parameter's value. If control of this dimension is to be perceptually based, a psychophysical relation between parameter value and brightness is needed. A psychophysical scaling of brightness as a function of parameter value may be obtained by playing timbres varying in the value of that single synthesis parameter and collecting subjective judgments of brightness for each. The resulting scale, stated in the form of a functional equation, can be inverted so that the parameter value associated with a desired brightness may be calculated directly.

Now most musicians are not going to want to design and execute carefully controlled psychophysical experiments every time they sit down to work on a new musical idea. But if the musician wants perceptual controls for a small number of timbral dimensions describing just the palette of timbres that will be used in a given piece of music, then a single session using *Palette* will suffice – and the whole session can run automatically. Furthermore, by including two extra steps into the psychophysical scaling procedure, an extremely useful body of information about a set of timbres may be generated. These three steps may be described briefly as follows:

1. **Sound Selection.** The user selects the range of timbres of interest for a given synthesis algorithm.
2. **Unidimensional Scaling.** The user generates a psychophysical scale for a single timbral dimension taken from that range of timbres.
3. **Timbre Matching.** The user matches the timbres resulting from other synthesis algorithms to those of the above algorithm on the scaled timbral dimension.

These three steps complete a scaling session whose results can be immediately implemented for the processing of musical scores (for example, musical event-list data processed within the *eled* program). A good example of a timbral dimension for which perceptually-based control is highly desirable is the synthetic instrument's dynamic character, especially if its dynamic character has been programmed to change in a naturalistic fashion with changes in pitch.

Example 1: Controlling dynamic character while varying pitch. The timbral qualities of an acoustic instrument change with the intensity and pitch at which the instrument is played. As an instrument is played with increasing intensity, the primary change in timbre that occurs may be in the brightness of the instrument's timbre, but the change may also include a number of other factors such as the perceived percussiveness of the instrument's attack. The manner in which this "dynamic character" of the instrument is expressed also changes with the pitch at which the instrument is played. Despite the fact that several factors may be changing with pitch and intensity, however, musicians can generally hear all the changes together as a global change in the dynamic character of the instrument. Synthetic timbres, on the other hand, have no intrinsic dynamic character, although a particular dynamic is usually programmed for (designed into) each. When musicians want to get control over the dynamic character of a number of synthetic voices over a particular pitch range, the *prv* (pitch-by-velocity) procedure in *Palette* is executed. A single synthesis parameter controlling dynamic character (key-velocity in the case of the DX7) is manipulated for each voice at several pitches covering the desired pitch range. Using three interactive tools, the *prv* procedure generates a psychoacoustic database which will allow the dynamic character of all the voices involved to be specified on a common subjective scale. What follows is an outline of a typical *prv* session.

Sound Selection. The musician specifies a single voice, for example, the voice named "marimba," from the set of voices to be scaled. Typing *prvadj marimba* puts the musician into a sound selection mode in which the pitch and velocity of the voice may be adjusted in real time. The adjustment is accomplished through the manipulation (via mouse) of a pointer displayed on a two-dimensional plot of pitch by velocity (illustrated on the left side of Figure 1). Depressing the left button on the mouse plays the voice at the current pitch and velocity of the pointer, very much as if one is moving about on a surface for "playing" the marimba (cf. Wessel, 1979). When the musician is satisfied with the pitch and velocity, depressing the right button records those values and places a marker on the screen at the current pointer position. The figure shows what the screen looks like after six settings have been recorded to demarcate the useful range of velocity as a function of pitch (how large and small the mallets – pictured on the right side of Figure 1 – can be at each pitch). Depressing the middle button terminates sound selection and writes the recorded pitch – velocity pairs into a data file that will be used in subsequent sessions.



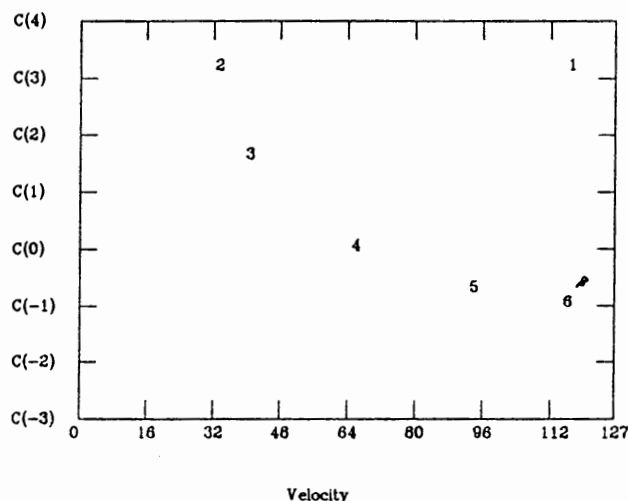


Figure 1. Sound selection: Musicians specify a region of useable timbres in a graph of pitch by key velocity.

Unidimensional Scaling. To generate a subjective scale for the perceived dynamic character of the marimba within the range of velocities specified in the sound selection session, the musician types *prvscl marimba*. The synthesizer then plays this voice at three pitches and five velocities selected from the range described in the data file written by *prvadj* (the three-by-five matrix of timbral stimuli is the default configuration). After the entire range of timbral stimuli has been heard, the musician enters a subjective-rating mode (examine Figure 2). The musician rates each timbre, presented in a random order, relative to the overall range of dynamic qualities present in the stimulus set. Subjective ratings are indicated by adjusting (via mouse) the length of a line segment along a track labeled from 1 to 9 (shown at the top-right of Figure 2). Once again, depressing the left button will allow the stimulus to be replayed until a satisfactory setting is made. This setting is recorded by depressing the right button and the musician proceeds to the next timbre-rating trial.

1 of 15
Adjust the line segment
The left button replays the stimulus
The right button records your response
Hit middle button to quit

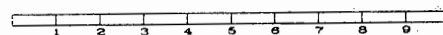


Figure 2. Unidimensional scaling: Musicians indicate subjective timbre ratings on a scale from 1 to 9 by adjusting the length of a line segment.



Figure 3. Timbre matching: Musicians adjust the value of one synthesis parameter to match the perceived character along one perceptual dimension between the comparison and the standard stimulus (see text).

When each of the fifteen stimuli have been rated, *prvscal* fits a curve to each set of five velocities and the associated subjective ratings to determine the shape of the psychophysical function that describes perceived dynamic character at each of the three pitches. The smooth curves fitted using the simplex algorithm (Caccci & Cacheris, 1984) usually account for more than 90% of the variance in the five points describing the curve. Typically, the five subjective ratings collected in the first session for each pitch will not be sufficient for obtaining useful curves, and so several iterations of the scaling procedure are executed. In each iteration, the results of the last curve fitting are used to predict how the synthesis parameter values should be spaced in the next so as to separate the timbres by roughly equal perceptual distances at each pitch. In effect, this adaptive scaling procedure collects subjective data for the stimulus levels that best determine the desired psychophysical function.

Timbre Matching. Having generated subjective scales for the dynamic character of one voice at a number of pitches, the musician chooses another voice for which similar controls are desired. By typing *prvmch marimba*, the musician enters the timbre-matching mode, in order to compare the two voices in the manner illustrated in Figure 3. The previously-scaled voice serves as a fixed reference (the standard) for the newly chosen voice (the adjustable comparison), and the musician matches the perceived dynamic of the comparison voice to each of a number of dynamic values of the standard that have been spaced at equal perceptual intervals. The procedure yields a common subjective scale for the two voices so that specifying a dynamic level of 5 (on the scale from 1 to 9) will result in the same perceived dynamic level for each. The matching procedure may be repeated for as many new voices as desired, the results always being referenced to the same standard scale.

Implementation. When the musician wants to use the resulting psychophysical relations to control the timbral synthesis specified in a musical score, the dynamic level (a value from 1 to 9) may be translated back into a velocity value in a pitch-dependent fashion by employing the inverse of the functions determined through unidimensional scaling (as simply as the pointing of the finger in Figure 4). Figure 5 shows the curves fitted to the perceived dynamic of the marimba voice for velocity variation at three different pitches. If the user specifies a dynamic of 7, nearly the same key-velocity (a value of 100) is required at the two lower pitches, G(0) and E(1). On the other hand, nearly maximum velocity (a value of 127) is required to give a similar perceived dynamic at the highest pitch, C#(2). On the subjective scale constructed for the dynamic character of the marimba, the lower-pitched tones will give subjective magnitudes up to a value of 10, but such magnitudes are unobtainable at the highest pitches.

At *NCM*, the translation of subjective values into usable synthesis data is implemented as a single command in the event list editing program *eled* (Decker & Kendall, 1985). Typing *prvtrans* within *eled* translates the value of the timbral descriptor (dynamic) into a value for the synthesis parameter (velocity) for all the appropriate notes in the score. Notes in the score which use timbres for which psychophysical data have not been collected are left unchanged (one is forced to specify these timbres in terms of synthesis parameters).

Choosing synthesis algorithms. Approaching the problem of developing a palette of timbres in the fashion described above requires a synthesizer that has already been programmed to give the kind of timbral variation the musician wants. Experience with scaling the dynamic character of DX7 voices that are distributed along with the synthesizer has shown that only about one voice in four has significant timbral variation with key-velocity. It is beyond the scope of this paper to offer suggestions regarding how one might go about reprogramming DX7 voices to get direct control over particular timbral features. Without proficiency in programming the synthesizer, the best one can do is sample all the voices available and choose those that have the desired responsiveness. In fact, one could argue for ignoring, to a large extent, the synthesis algorithms that underly the timbral palette one is developing. Ideally, it should be possible to choose synthesis algorithms strictly in terms of their perceptual features without even looking at how an algorithm has been programmed for a given voice.

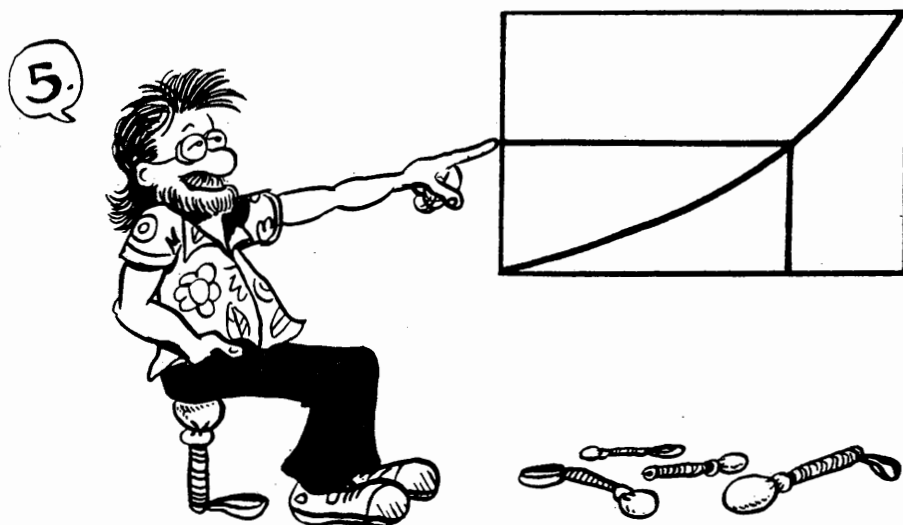


Figure 4. Implementation: Musicians can specify timbres along the same subjective scale used above for recording perceptual judgments. The curve describes the psychophysical relation between velocity (abscissa) and perceived dynamic character (ordinate) at a given pitch.

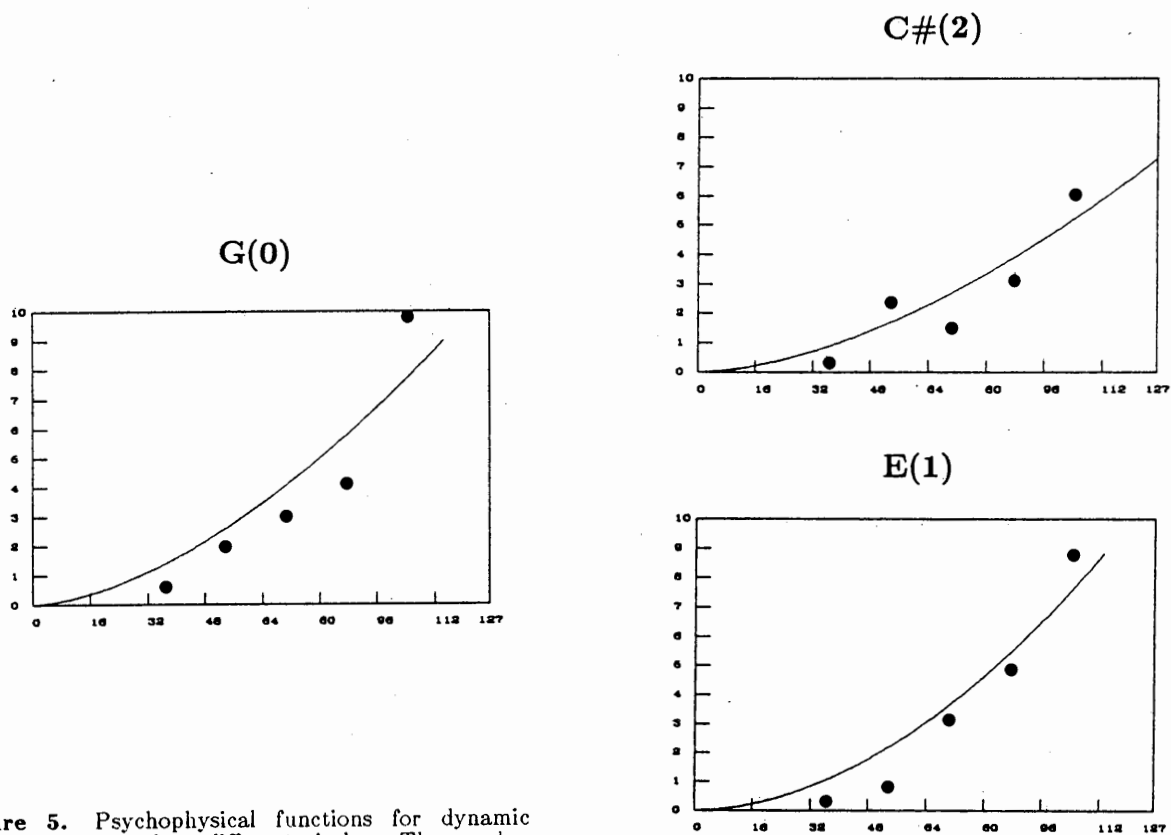


Figure 5. Psychophysical functions for dynamic versus velocity at three different pitches. The number in parentheses indicates the octave relative to middle C in which the pitch is located.

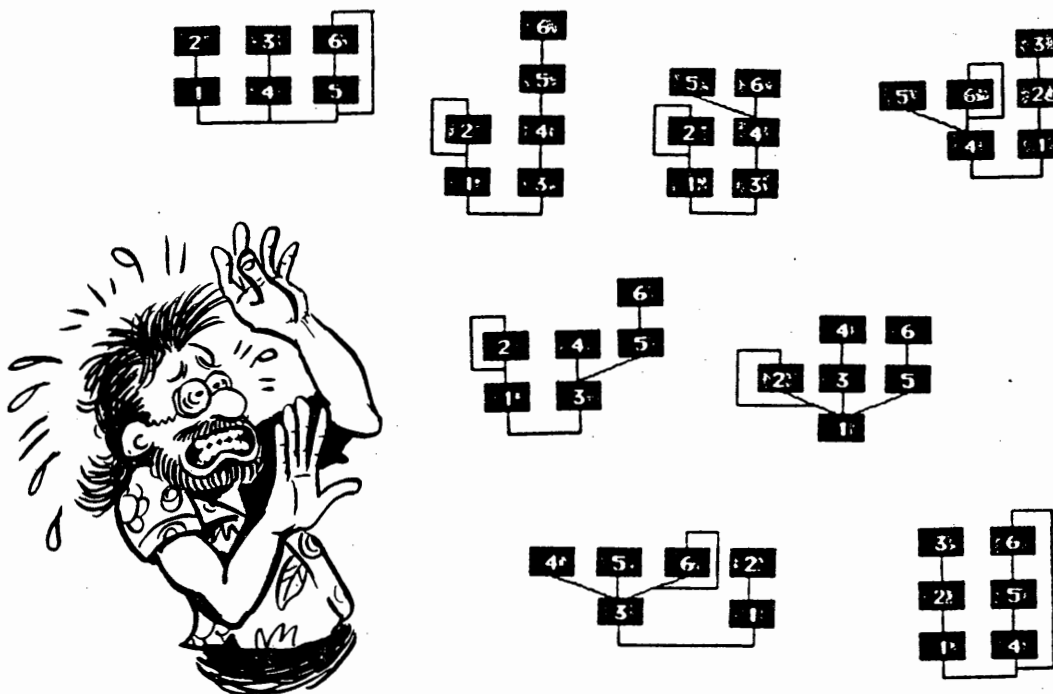


Figure 6. Choosing synthesis algorithms: a few examples of the FM synthesis algorithms available on the Yamaha DX7 are pictured. Each box is an operator (combined oscillator and envelope generator) that may be used to modulate the frequency of the operator connected directly below it.

The DX7 synthesizer makes available 32 algorithms of six operators which may be programmed in terms of well over 100 parameters. (Figure 6 shows 8 of the DX7 frequency-modulation synthesis algorithms. For an explanation of the basis of the synthesis technique, see Chowning, 1973; Schottstaedt, 1977.) In the face of this sort of complexity, a reasonable approach to determining the potential of the synthesizer would be to systematically examine all the perceptual dimensions through which the DX7 timbres move. Beyond the preset voices that have been provided for the Yamaha DX7, there is a multidimensional "timbre space" that could be explored (as is suggested in Figure 7). Without paying any attention to the synthesis algorithms underlying a palette of timbres, we can learn a lot about how to organize the timbres upon that palette from an analysis of their perceived similarities. This purely psychological approach (Shepard, 1980) has some advantages over the psychophysical approach described above, especially for complex, naturalistic timbres for that the relevant physical dimensions are as yet poorly understood. Furthermore, this perceptual exploration of timbre provides an important body of information for the *Palette* user which complements the results of the psychophysical scaling stored in the psychoacoustic database. The "timbre space" revealed through this approach provides a more global picture of the musician's timbral palette; and the perceptual dimensions determined for the set of timbres can be used as a "control structure" for timbral manipulations (Wessel, 1979).

3. EXPLORATION OF PERCEIVED TIMBRE

By asking a musician the direct question of how similar to each other two timbres sound, a measure of the perceptual distance between the two timbres may be obtained. Regardless of how a set of timbres is synthesized, a matrix of perceptual distance data can be generated for the set by making similarity judgments for all pairwise comparisons of the timbres within the set (using the process illustrated in Figure 8). The matrix may be submitted to a number of complementary forms of analysis, the most common of which are multidimensional scaling, tree-fitting, and clustering (Shepard, 1982). These analyses operate upon the inter-stimulus distances in order to capture some aspect of the perceptual relationship between the stimuli. Each can be used to construct a geometric and/or graphical representation of the psychological structure underlying the perception of the timbres within the set of sound stimuli.

The concept of exploring the perceptual space describing a set of timbres through multidimensional scaling analysis was brought to the attention of the computer music community through the work of John Grey (eg., 1975) and Dave Wessel (eg., 1979). In these studies, observers were asked to rate the perceived similarity of two timbres in relation to all the pairs of timbres presented in a listening session. After all pairwise comparisons of the timbres within the set were made, the similarity judgments were treated as measures of the perceptual distance between the tim-



Figure 7. Exploring timbre space.

— conductor, '83

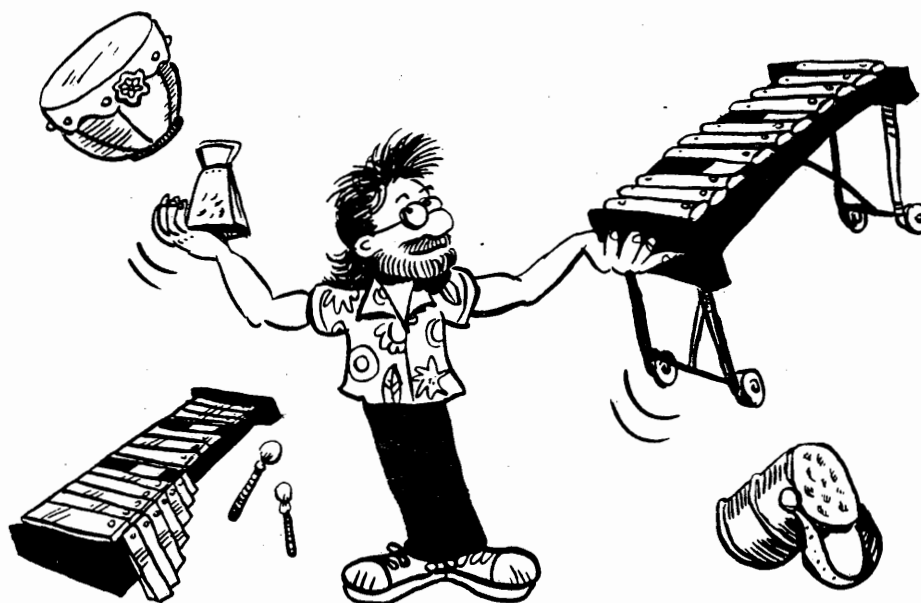


Figure 8. Timbre comparison: Musicians rate the perceptual similarity for each pair of timbres taken from a set of timbres.

bres, and each timbre could then be treated as a point in a perceptual space. Multidimensional scaling analysis determined the spatial position for each timbre which best fit the perceptual distances between them. Grey (1975) also submitted his perceptual distance data to cluster analysis to provide additional insight into the perceptual organization within the set of timbres. He

superimposed the hierarchical clustering solution for a set of timbres upon the spatial map obtained through MDS analysis. A non-spatial alternative to combined MDS and cluster analysis that yields a picture quite useful to the musician's exploration of perceived timbre is the tree-fitting analysis developed by Sattath & Tversky (1977).

Voice name	harp_1	banjo	jazz_guit	koto	lute	clas.guit	e.bass_1	pluck_bass
harp_1	10.00	7.78	5.29	8.55	4.74	7.93	6.50	4.45
banjo	8.69	10.00	2.50	8.20	6.84	2.16	2.20	2.52
jazz_guit	5.62	2.50	10.00	3.56	3.45	7.92	3.72	4.54
koto	8.41	6.81	5.83	10.00	7.61	2.83	1.37	4.33
lute	9.03	5.05	8.00	6.25	10.00	3.09	6.81	3.92
clas.guit	4.92	0.40	2.88	1.41	1.94	10.00	4.53	7.35
e.bass_1	7.65	1.53	2.94	1.42	1.95	8.32	10.00	5.96
pluck_bass	4.52	1.52	6.80	0.70	1.56	8.16	7.76	10.00

Table 1. Similarity judgments for all pairwise comparisons of eight of the DX7's "plucked" timbres.

Example 2: Tree-fitting for eight plucked timbres. The author's similarity judgments for all pairwise comparisons of eight "plucked" timbres produced by the DX7 are shown in Table 1. The full matrix of similarity judgments was constructed by presenting each comparison twice, with the order of the sound stimuli reversed in the second presentation. Some of the pairs produced fairly different judgments depending on which timbre was presented first, but the overall effects of presentation order proved negligible. The matrix was symmetrized through averaging so as to cancel out this order effect for each timbre pair. When submitted to additive tree-fitting analysis using the ADDTREE program (Corter, 1982), these data resulted in the tree structure shown in Figure 9. The tree represents not only the perceptual distance between each pair of timbres (which is captured by the distance one must move along the tree to get from one timbre to another), but also the perceptual organization of the timbres. Each node of the tree (where branches join) may be treated as a timbral category which represents the collective character of all timbres that lie beyond that node in the tree (away from the central root of the tree). For example, the DX7 voices named after the electric guitar, the plucked bass, and the classical guitar are all grouped together, probably because they all have a "flabby" plucked sound. The remaining timbres all have a "cleaner" plucked sound, but two - the banjo and the koto - have more of a "twang" than the others. In general, the

results of these additive tree fittings usually confirm the musician's expectations regarding how a set of timbres might be organized. Nonetheless, these results are very useful for the compositional organization of timbre in that they provide an explicit representation of perceptual relations that are usually only intuitively grasped.

Now you could ask musicians to categorize a number of timbres and name the features used in these categorizations. However, only if the timbral categories have clear verbal labels will this task be relatively easy. For example, the dimension extending from "clank" to "clunk" might be described as "clunkiness," but this term has no meaning outside of the context of the stimulus set for which it was generated. There is an additional problem that may make such judgments more difficult. In order for timbres to be so categorized, the dimensions along which the timbres differ must be analyzable - that is, distinct enough from one another that the rating of a timbre on one dimension will not be influenced by the value of that timbre along the other dimensions. The interdependence of timbral dimensions typically precludes such ease in generating subjective reports. Whereas the color of a polygon along a scale from blue to green to yellow will not be significantly influenced by whether the polygon is a square or a triangle, the position of a timbre along the dimension extending from "plucked" to "struck" will likely interfere with subjective judgments of percussiveness.

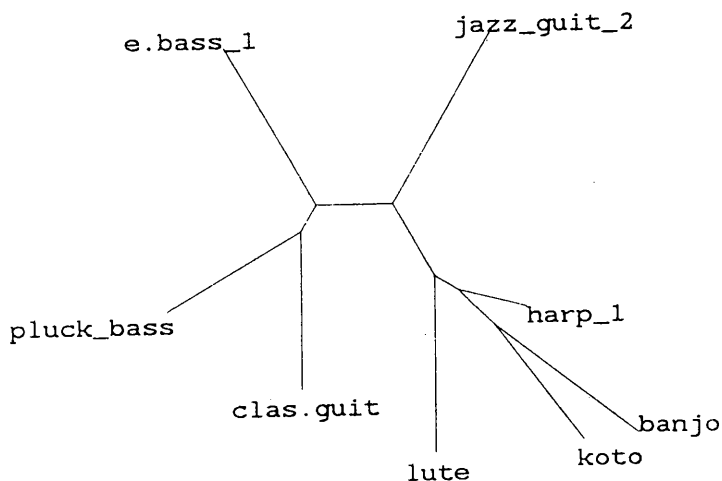


Figure 9. Additive tree for 8 "plucked" timbres: Results of a tree-fitting analysis on similarity judgments for one observer.

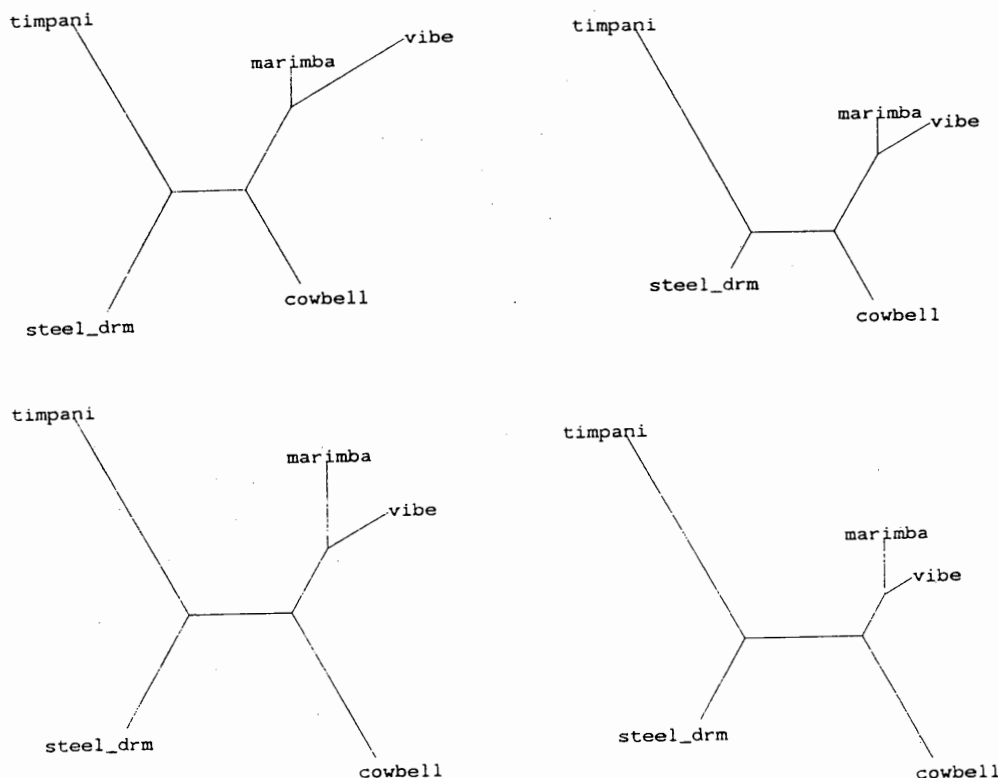


Figure 10. Additive trees for 5 "struck" timbres:
Results of tree-fitting analyses on similarity judgments
for four observers.

A question that must be raised when exploring perceived timbre on the basis of these direct judgments concerns the generality of the results of tree-fitting analysis. Will musicians agree on perceptual relations between timbres? The answer is both yes and no. If musicians are asked simply to rate similarity without paying attention to any particular detail, they generally produce results that agree. Figure 10 shows the additive trees for five "struck" timbres based on the similarity judgments of four listeners. Note that the clustering of timbres is identical across all listeners, though timbres are placed at differing distances from each branching node of the tree by each listener. On the other hand, when listening to a set of timbres that are to be used in a particular composition for a particular effect, a musician may want to focus on that timbral aspect within the set of timbres that will serve a musical function in that composition. Under these circumstances, the musician may focus on subtleties that others might ignore, and may even choose not to pay attention to the most salient timbral dimensions varying within the set of timbres. Here, the results will make sense only in the musical context the musician had in mind, but the results will provide a representation that can aid the musician's timbral exploration for composition.

4. THE PALETTE WORKING ENVIRONMENT

What follows is a description of the software features and information flow that compose the *Palette* working environment, as illustrated in Figure 11 (the cloud issuing lightning indicates the primary function of *Palette*, which is to provide the link between psychological reality and computer representation). This is a general overview of the software package still under development.

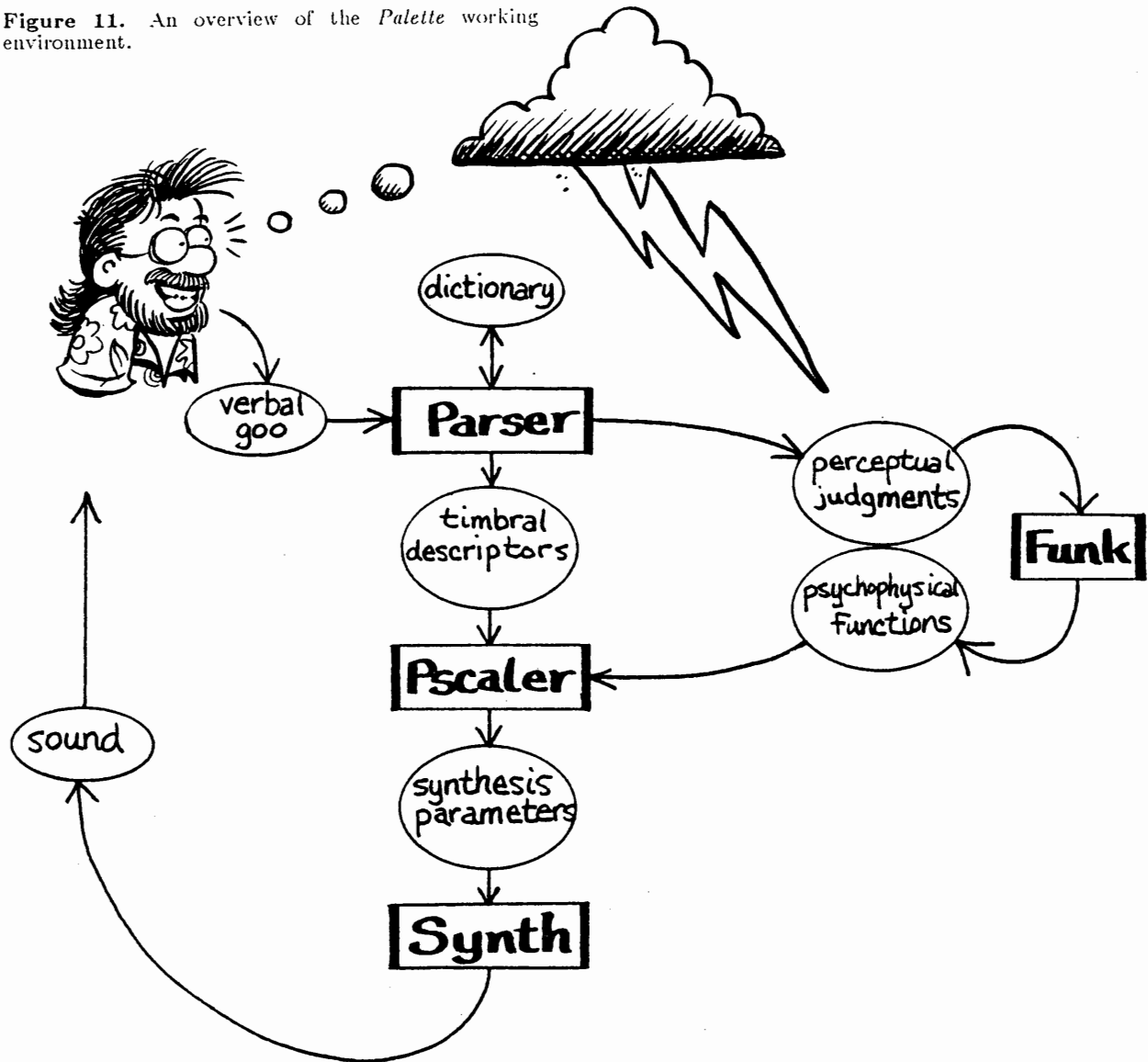
Parser accepts verbal descriptions as well as numerical judgments from the user (generically, verbal goo, owing to its fundamentally unstructured character). Using a database full of verbal commands, objects, and modifiers (a dictionary), the *Parser* creates outputs of two forms:

1. **Timbral descriptors**, which are terms which describe simple features of the perceptual experience for a given synthetic timbre, and
2. **Perceptual similarity judgements**, which are subjective ratings of the relationship between two timbres presented for pairwise comparison.

Pscaler (*psychophysical scaler*) is a software filter that accepts the first sort of *Parser* output (timbral descriptors) and translates them into synthesis parameter values according to a set of user-individualized psychophysical functions. Note that synthesis parameters need not have a one-to-one relationship to timbral descriptors. In fact, it is likely to be the case that each synthesis parameter will depend on many of the timbral descriptors passed down from the *Parser* to *Pscaler*.

Synth is the synthesis algorithm (hardware or software synthesis) to be used, which is chosen from a number of alternative algorithms depending on the timbral palette envisioned by the user. In the case of software synthesis, the algorithms available are sufficiently flexible to allow for experimentation within the algorithm itself, while efficiency is maintained through the use of software switches that disable those algorithmic manipulations not desired. Other algorithms may be installed into *Palette*, but they must adopt the standardized data structure that *Pscaler* uses in communicating with the *Synth* module. In the case of hardware synthesis, such as MIDI-controlled DX7, the program-

Figure 11. An overview of the *Palette* working environment.



ming of the synthesizer is achieved via special-purpose software such as *NCM's dzed* program written by Brian Schmidt.

Funk does functional analysis on the perceptual similarity judgments that *Parser* has placed into a database of subjective ratings (stored in parallel with the synthesis parameter values associated with those judgments). The functional equations fit to these data are placed into a database to which the *Pscaler* has access.

5. CONCLUSION

Methods for obtaining perceptual controls over synthesizers that have proved useful to musicians working in a real-time computer music environment have been outlined. Though still in an early stage of its development, the *Palette* software package is achieving its goal of allowing musical timbre to be described in strictly psychological terms, relieving the composer of the responsibility for engineering sounds at the level of the synthesis algorithm itself. This has been accomplished by shifting the focus of psychoacoustic investigation, from answering questions about

how timbre is perceived in general to solving the problem of how a given musician perceives particular timbral modifications (limited to the set of timbres to be used in a particular composition). It is hoped that these procedures for controlling timbral synthesis will spread outside current applications at *NCM* and find application to a number of related problems in musical composition and production.

6. ACKNOWLEDGMENTS

I am grateful to all the staff and students at *NCM* for extremely helpful discussion and criticism. I would especially like to thank Gary Kendall, Shawn Decker, and Brian Schmidt for their time, effort, and commitment to the development of the *Palette* software package. A lot of credit is also due to Derek Ludwig for help with the graphics programming, including the program that constructed the tree diagrams from the output of the ADDTREE program which he wrote for me at a moment's notice. Thanks to John Goodrich for the cartoon artistry and the creation of the *Palette* cartoon musician.

REFERENCES

- Caccci, M. S., & Cacheris, W. P. (1984) Fitting curves to data. *Byte*, May, 340-362.
- Corter, J. E. (1982) ADDTREE/P: A PASCAL program for fitting additive trees based on Sattath and Tversky's ADDTREE algorithm. *Behavior Research. Methods, & Instrumentation*. 14, 353-354.
- Decker, S. L., & Kendall, G. S. (1985) A unified approach to the editing of time-ordered events. *Proceedings of the 1985 International Computer Music Conference*, Vancouver.
- Grey, J. M. (1975) An exploration of musical timbre. Ph.D. dissertation (Department of Music Report No. STAN-M-2, Stanford University) (unpublished).
- Risset, J. C. & Mathews, M. V. (1969) Analysis of musical instrument tones. *Physics Today*. 22, 23-30.
- Roth, J. M., Kendall, G. S., & Decker, S. L. (1985) A versatile soundfile system for UNIX. *Proceedings of the 1985 International Computer Music Conference*, Vancouver.
- Sattath, S., & Tversky, A. (1977) Additive similarity trees. *Psychometrika*. 42, 319-345.
- Schottstaedt, B. (1977) The simulation of natural instrument tones using frequency modulation with a complex modulating wave. *Computer Music Journal*. 1, 46-50.
- Shepard, R. N. (1980) Multidimensional scaling, tree-fitting, and clustering. *Science*. 210, 390-398.
- Wessel, D. L. (1979) Timbre space as a musical control structure. *Computer Music Journal*. 3, 45-52.

